AI 101 for Mechanical Engineers · Module 2

Different questions need different outputs

Lesson 2.2

What you will learn

> Match an engineering question to the required output — prediction, forecasting, diagnosis, control or optimization — and distinguish ML training optimization from engineering design optimization.

Lesson overview

Module 2 · Lesson 2

Different questions need different outputs

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1 Match the task to the question

Task	What it does	Lesson example
Prediction	Estimates an unknown output from supplied inputs	Estimate pump power at a stated operating condition
Forecasting	Estimates a future outcome	Estimate tomorrow's HVAC demand or PV energy
Diagnosis	Identifies which condition or cause explains the evidence	Investigate possible bearing damage
Control	Selects actions over time to regulate a system	Adjust fan speed to maintain a target temperature
Optimization	Searches feasible choices to improve an objective within constraints	Choose insulation thickness to reduce life-cycle cost

Forecasting uses only information available when the forecast is made.

An unusual vibration alert needs investigation. Confirming damage requires further evidence, such as inspection.

2 Two uses of optimization

ML training

Change model parameters → Reduce training loss

Example: adjust model weights.

Engineering design or operation

Change design or operating variables → Improve an engineering objective

Must meet the stated physical constraints.

Before optimizing, state:

1. What can change?
2. What should improve?
3. What limits must be met?

Predicting pump power does not, by itself, optimize pump operation.

Optimization also needs decision variables, an objective, constraints and a search.

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Prediction and forecasting estimate different things

Prediction

- Estimates unknown output from supplied inputs
- e.g. pump power at a stated condition

Forecasting

- Estimates a future outcome [18]
- e.g. tomorrow’s HVAC demand or PV generation

A forecast must use only information available when it is issued.

An anomaly alert is not a diagnosis

- Diagnosis identifies health state, fault type or location [10]
- Unusual vibration can produce an anomaly alert
- Confirming bearing damage needs more evidence, e.g. physical inspection

Control acts over time; optimization searches choices

Control

- Selects actions to regulate behaviour over time [7]
- e.g. adjust fan speed to maintain target temperature

Optimization

- Searches feasible choices to improve an objective [19]
- Must satisfy constraints
- e.g. insulation thickness to reduce life-cycle cost

> A power prediction can inform the search, but prediction alone does not optimize.

“Optimization” has two different uses

ML training

- Changes internal model parameters, such as weights
- Reduces a mathematical loss function [4]

Engineering design

- Changes physical design or operational variables
- Improves an engineering objective under physical constraints

> Identify the variables, objective and limits before using the word “optimization”.

Optimization needs more than a prediction

1. Decision variables: what can be changed?
2. Objective: what should improve?
3. Constraints: which limits must hold?
4. Search over the feasible choices

CHECK YOUR UNDERSTANDING

Why does predicting pump power not itself optimize pump operation?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why does predicting pump power not itself optimize pump operation?

> Prediction estimates power for supplied inputs. Optimization also needs decision variables, an objective, constraints and a search over feasible choices.

CHECK YOUR UNDERSTANDING

Does an unusual vibration alert confirm bearing damage?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does an unusual vibration alert confirm bearing damage?

> No. It indicates that investigation is required; diagnosis needs additional evidence.

Key takeaways

- Prediction estimates an output; forecasting estimates a future output
- Diagnosis identifies the condition; control selects regulating actions
- Optimization needs objectives, constraints and feasible decision variables
- Reducing training loss differs from improving engineering performance

References

[4] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA, USA: MIT Press, 2016.

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[18] R. J. Hyndman and G. Athanasopoulos, Forecasting: Principles and Practice, 3rd ed. Melbourne, Australia: OTexts, 2021. [Online]. Available: https://otexts.com/fpp3/

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Numbers match the course reference list. The written lesson is the authoritative text.